C231

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* tarea mostrar que $\hat{\theta}_1$ y $\hat{\theta}_2$ son insesgados.

* $f(y) = \frac{1}{\theta}, \quad 0 \le y \le \theta$ $\qquad F(y) = \begin{cases} 0, & \text{si } y < 0 \\ \frac{y}{\theta}, & 0 \le y \le \theta \\ 1, & \text{si } y > \theta \end{cases}$

* $$\begin{aligned} var(\hat{\theta}_1) &= var(2\bar{Y}) \\ &= 4\, Var(\bar{Y}) \\ &= 4 \frac{\sigma^2}{n} \\ &= \frac{4}{n} \cdot \frac{\theta^2}{12} \\ &= \frac{\theta^2}{3n} \end{aligned}$$

$$\left\{ \; var(Y_i) = \sigma^2 = \frac{(b-a)^2}{12} = \frac{(\theta-0)^2}{12} = \frac{\theta^2}{12} \right.$$

* $$\begin{aligned} var(\hat{\theta}_2) &= var\left(\frac{n+1}{n} Y_{(n)}\right) \\ &= \left(\frac{n+1}{n}\right)^2 Var\left(Y_{(n)}\right) \end{aligned}$$

$$Var(Y_{(n)}) = E(Y_{(n)}^2) - E(Y_{(n)})^2$$

$$E(Y_{(n)}) = \int_0^\theta y f_{(n)}(y)\, dy$$

$$\begin{aligned} f_{(n)}(y) &= n\,[F(y)]^{n-1} f(y) \\ &= n \left[\frac{y}{\theta}\right]^{n-1} \frac{1}{\theta} \\ &= n \frac{y^{n-1}}{\theta^{n-1}} \cdot \frac{1}{\theta} \\ &= \frac{n y^{n-1}}{\theta^n} \end{aligned}$$

$$\begin{aligned} E(Y_{(n)}) &= \int_0^\theta \frac{y \cdot n y^{n-1}}{\theta^n}\, dy \\ &= \frac{n}{\theta^n} \int_0^\theta y^n\, dy \\ &= \frac{n}{\theta^n} \left.\frac{y^{n+1}}{n+1}\right|_0^\theta = \frac{n}{\cancel{\theta^n}} \cdot \frac{\theta^{\cancel{n}+1}}{n+1} \\ &= \frac{n\theta}{n+1} \end{aligned}$$

$$\begin{aligned} E(Y_{(n)}^2) &= \int_0^\theta \frac{y^2\, n y^{n-1}}{\theta^n}\, dy \\ &= \frac{n}{\theta^n} \int_0^\theta y^{n+1}\, dy \\ &= \left(\frac{n}{n+2}\right) \theta^2 \end{aligned}$$

$$\begin{aligned} V(Y_{(n)}) &= \left(\frac{n}{n+2}\right)\theta^2 - \left(\frac{n\theta}{n+1}\right)^2 \\ &= \theta^2 \left[\frac{n}{n+2} - \left(\frac{n}{n+1}\right)^2\right] \end{aligned}$$

$$\begin{aligned} V(\hat{\theta}_2) &= Var\left(\frac{n+1}{n} Y_{(n)}\right) \\ &= \left(\frac{n+1}{n}\right)^2 V(Y_{(n)}) \\ &= \left(\frac{n+1}{n}\right)^2 \theta^2 \left[\frac{n}{n+2} - \left(\frac{n}{n+1}\right)^2\right] \\ &= \frac{\theta^2}{n(n+2)} \end{aligned}$$

Comprobar.

$$eff(\hat{\theta}_1, \hat{\theta}_2) = \frac{Var(\hat{\theta}_2)}{Var(\hat{\theta}_1)} = \frac{\frac{\cancel{\theta^2}}{\cancel{n}(n+2)}}{\frac{\cancel{\theta^2}}{3\cancel{n}}} = \frac{3}{n+2}$$

Si $n > 1$, la $var(\hat{\theta}_1) > var(\hat{\theta}_2)$ por tanto nos quedamos con $\hat{\theta}_2$.

$E(\bar{Y}) = \mu$, ya sabemos que $\bar{Y}$ es insesgado.

$Var(\bar{Y}) = \frac{\sigma^2}{n}$

$$\lim_{n\to\infty} var(\bar{Y}) = \lim_{n\to\infty} \frac{\sigma^2}{n} = 0$$

$\bar{Y}$ es un estimador consistente para μ.

$$E(\hat{\theta}) = E(Y/n) = \frac{1}{n}E(Y) = \frac{1}{n}(np) = p \; ✓$$

$$Var(\hat{\theta}) = \frac{1}{n^2} Var(Y) = \frac{1}{n^2} \cdot npq = \frac{1}{n} pq$$

$$\lim_{n\to\infty} Var(\hat{\theta}) = \lim_{n\to\infty} \frac{1}{n} pq = 0$$

$\hat{\theta} = \frac{Y}{n}$ es consistente para p.

$$\begin{aligned} L(\theta) &= L(y_1, \ldots, y_n \mid \theta) \\ &= f(y_1, \ldots, y_n \mid \theta) \\ &= f(y_1 \mid \theta)\, f(y_2 \mid \theta) \cdots f(y_n \mid \theta) \\ &= \frac{1}{\theta} e^{-y_1/\theta}\, \frac{1}{\theta} e^{-y_2/\theta} \cdots \frac{1}{\theta} e^{-y_n/\theta} \\ &= \underbrace{\left(\frac{1}{\theta}\right)^n e^{-\frac{1}{\theta} n\bar{y}}}_{g(u,\theta)} \; \underbrace{(1)}_{h(y_1, \ldots, y_n)} \end{aligned}$$

$U = \bar{Y}$ es un estadístico suficiente para θ.